



FSK116 – Announcement

Sem Test Results: This Week

Tut Test 9: YES!

Sick Test: You'll receive email by about Thurs

Online Test: mid to end of next week

EXAM: 19th July, 12h00 to 15h00



Simple Harmonic Motion

Ch 15 – Oscillations



Springs

Energy

Pendulums



Simple Harmonic Motion (SHM)

▶ Periodic motion or harmonic motion

- motion that repeats itself at regular intervals.
- Particle/object moving back and forth repeatedly.

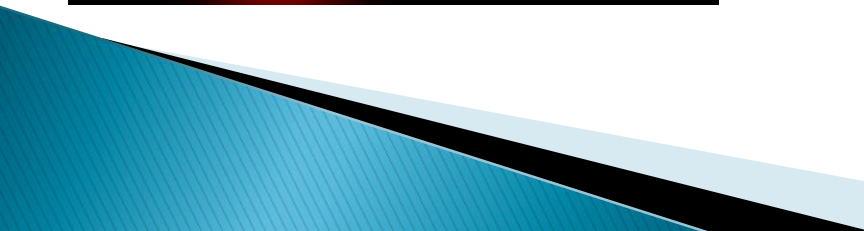
▶ Frequency f

- number of completed oscillations per second.
- SI unit: 1 hertz (Hz) = 1 oscillation/s
= 1 (s⁻¹)

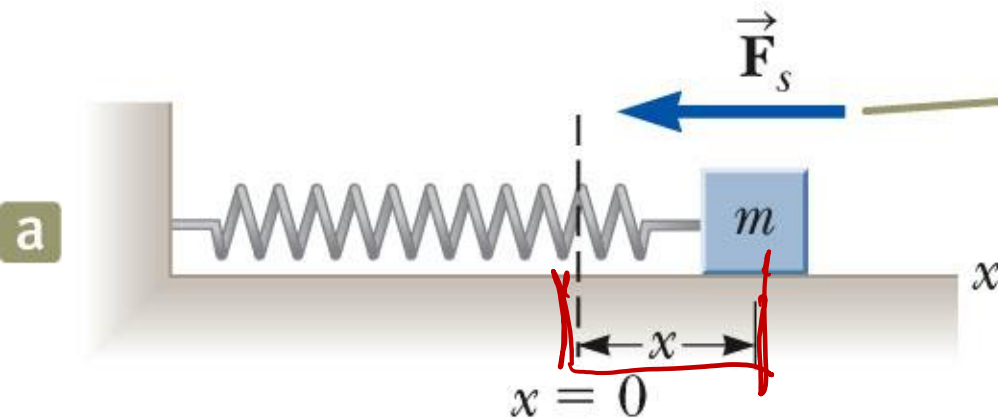
▶ Period T

- time for one complete oscillation.
-

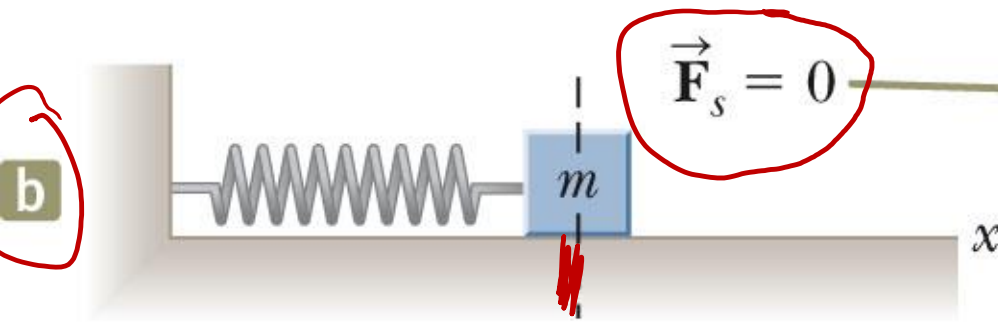
$$T = \frac{1}{f}$$



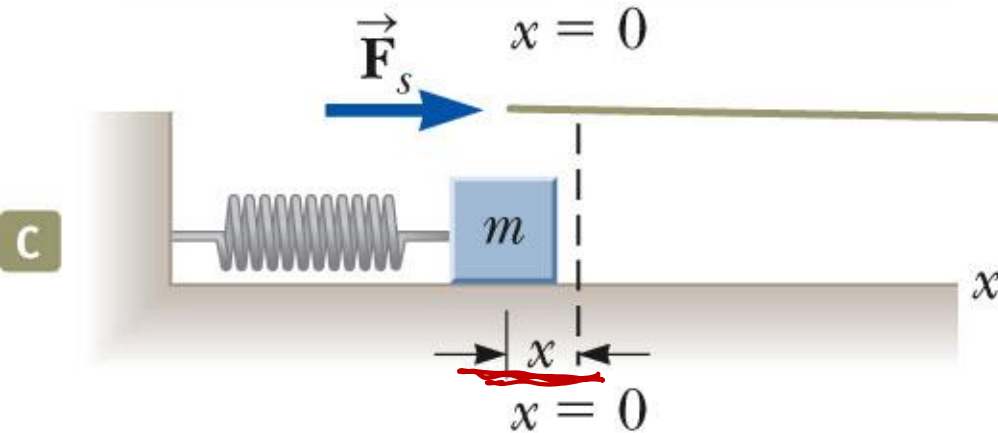
Motion of mass-spring system



When the block is displaced to the right of equilibrium, the force exerted by the spring acts to the left.



When the block is at its equilibrium position, the force exerted by the spring is zero.



When the block is displaced to the left of equilibrium, the force exerted by the spring acts to the right.

Motion of mass-spring system

- ▶ Block of mass, m , attached to the end of a spring.
- ▶ Block moves back and forth on a frictionless horizontal surface.

▶ For a spring we have a restoring force: $F = -kx$

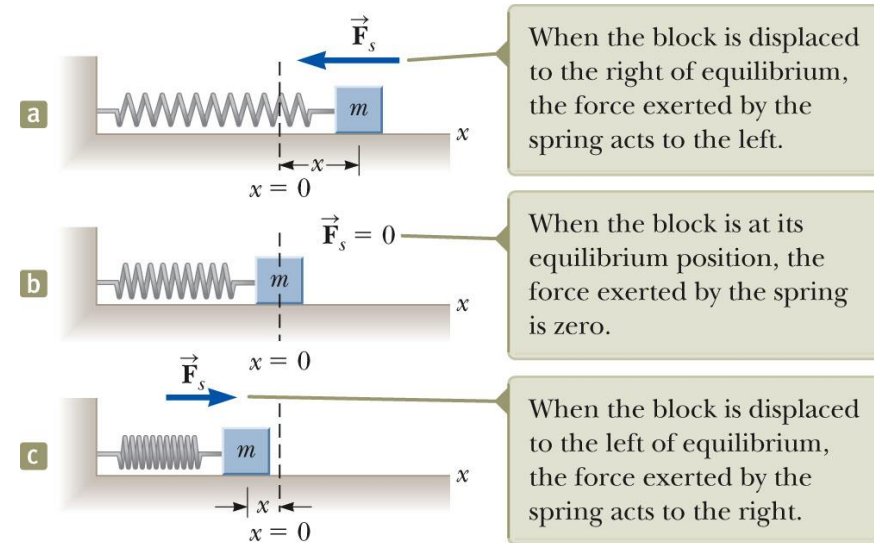
▶ But also : $\sum F_x = ma_x$ N II

▶ So $a_x = -\left(\frac{k}{m}\right)x$

$a \propto -x$

- ▶ Systems with this behaviour are said to exhibit

SIMPLE HARMONIC MOTION



Simple Harmonic Motion (SHM)

- From previous: $a_x = -\frac{k}{m}x$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

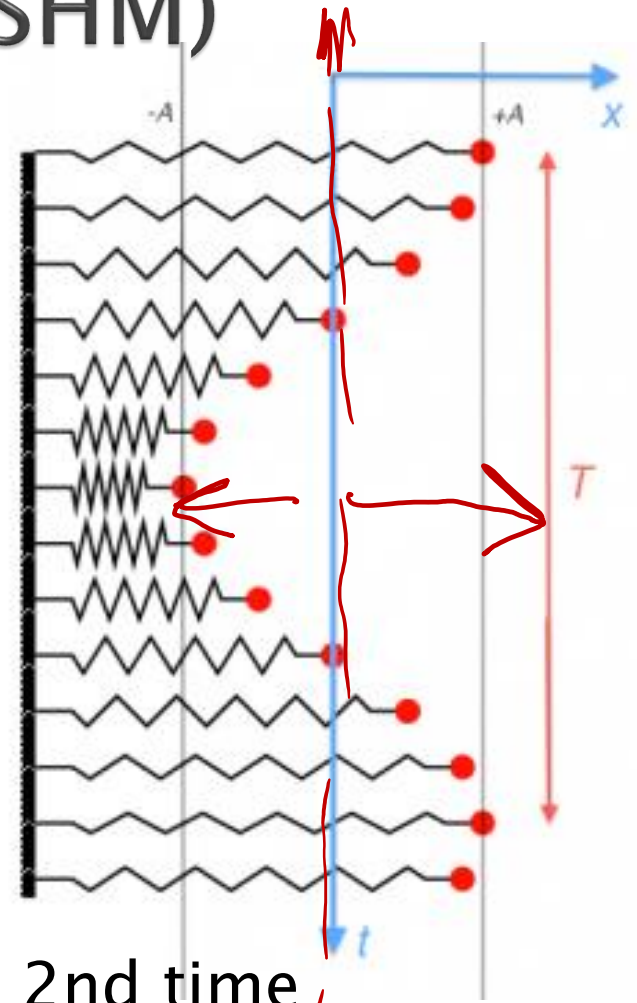
- We define:

$$\omega^2 \equiv \frac{k}{m}$$

- Then:

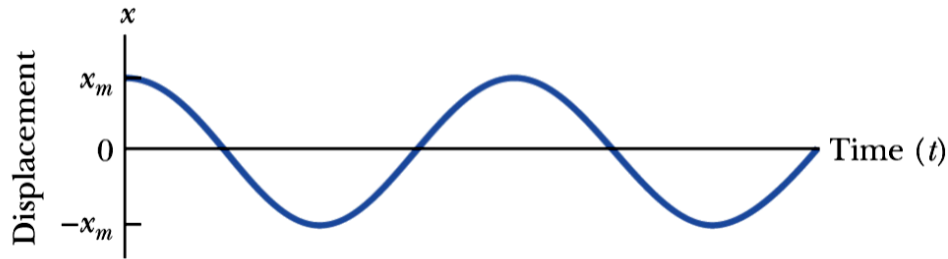
$$\frac{d^2x}{dt^2} = -\omega^2 x$$

- We need to find a function $x(t)$ whose 2nd time derivative is the negative of the original function.
- Looks like a sinusoidal function.



Simple Harmonic Motion (SHM)

$$x(t) = A \cos(\omega t + \phi)$$



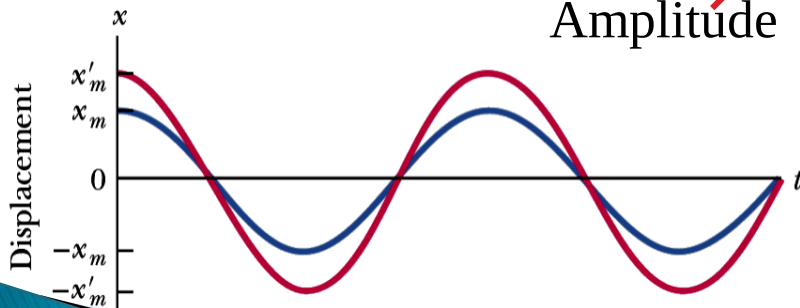
Displacement at time t

$$x(t) = A \cos(\omega t + \phi)$$

Amplitude

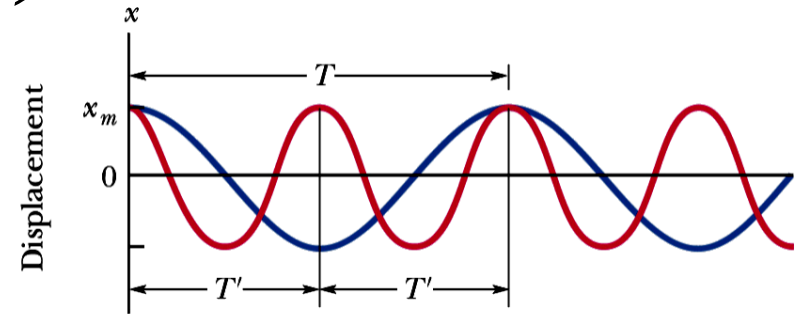
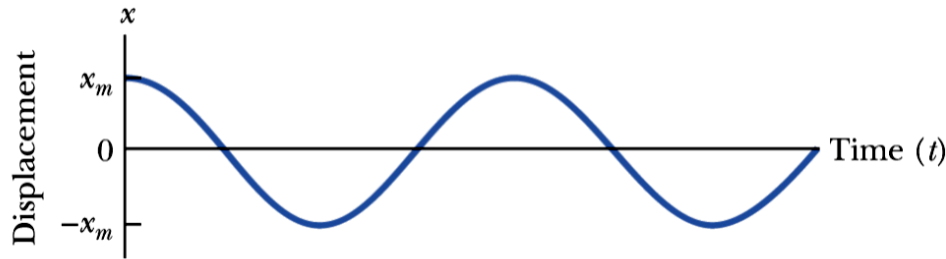
Time

Phase



Simple Harmonic Motion (SHM)

$$x(t) = A \cos(\omega t + \phi)$$



$$\omega = \frac{2\pi}{T} = 2\pi f$$

Displacement at time t

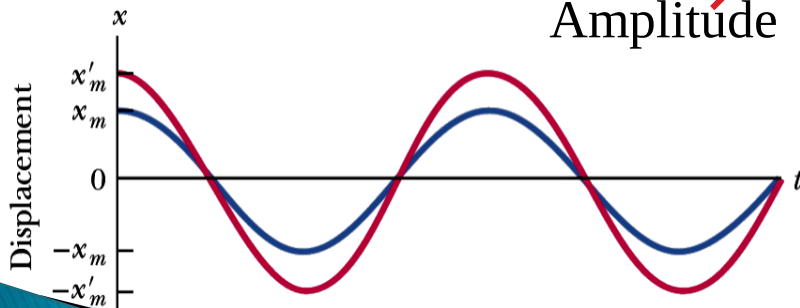
Angular frequency

$$x(t) = A \cos(\omega t + \phi)$$

Amplitude

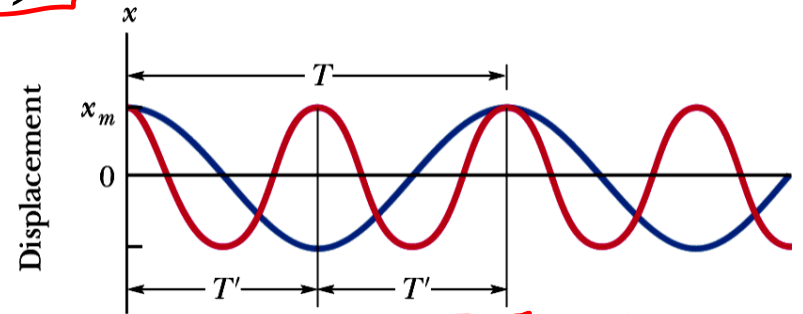
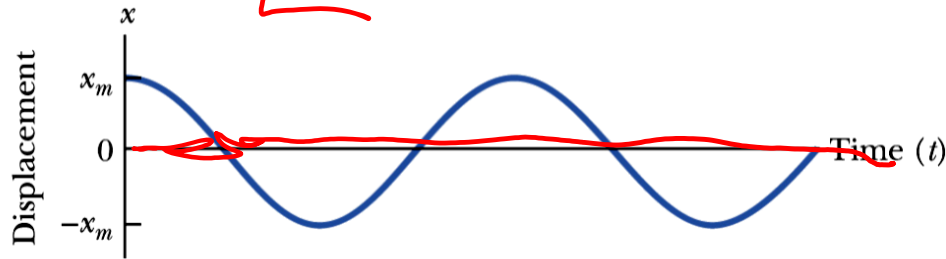
Time

Phase



Simple Harmonic Motion (SHM)

$$x(t) = A \cos(\omega t + \phi)$$



$$\omega = \frac{2\pi}{T} = 2\pi f$$

Displacement at time t

Angular frequency

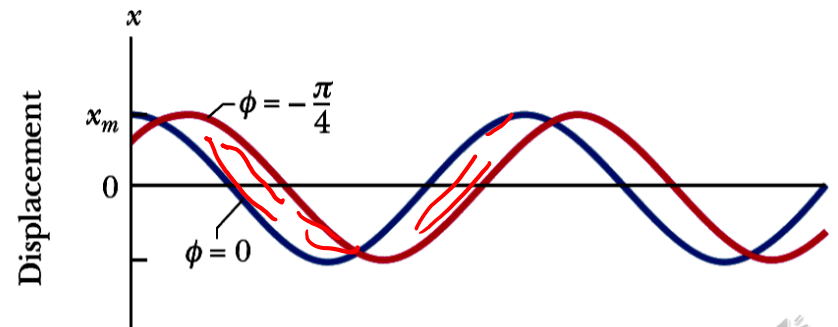
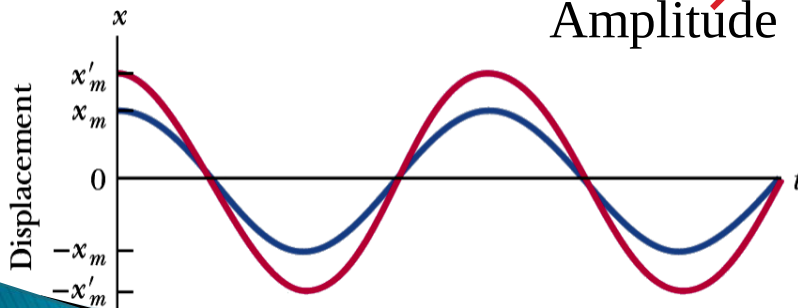
$$x(t) = A \cos(\omega t + \phi)$$

Phase constant /
Phase angle

Amplitude

Time

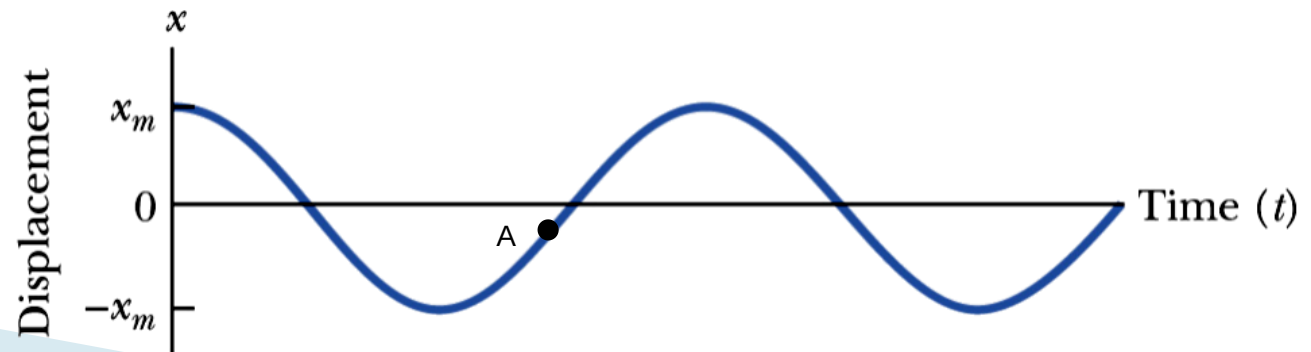
Phase



Problem

Consider the graphical representation of SHM as described mathematically by the equation we have just looked at. When the particle is at point A, what can you say about the particle's position and velocity?

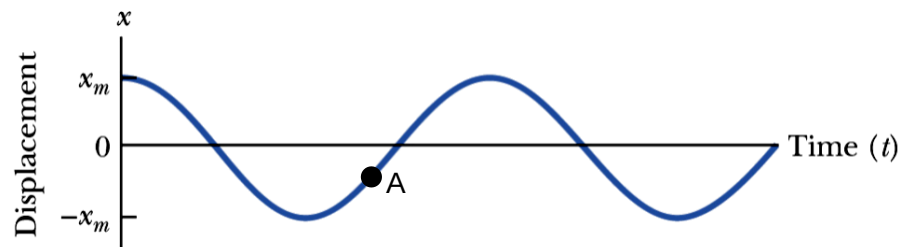
- a) The position and velocity are both positive.
- b) The position and velocity are both negative.
- c) The position is positive and velocity is zero.
- d) The position is negative and velocity is zero.
- e) The position is negative and velocity is positive.
- f) The position is positive and velocity is negative.



Problem

Consider the graphical representation of SHM as described mathematically by the equation we have just looked at. When the particle is at point A, what can you say about the particle's position and velocity?

- a) The position and velocity are both positive.
- b) The position and velocity are both negative.
- c) The position is positive and velocity is zero.
- d) The position is negative and velocity is zero.
- e) The position is negative and velocity is positive.**
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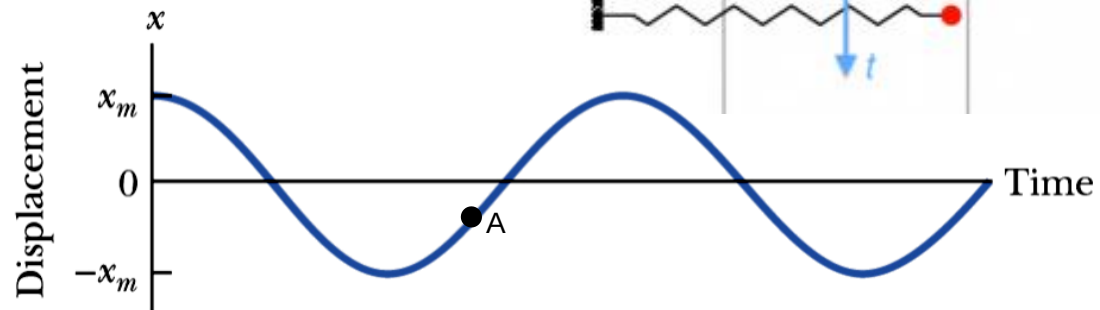
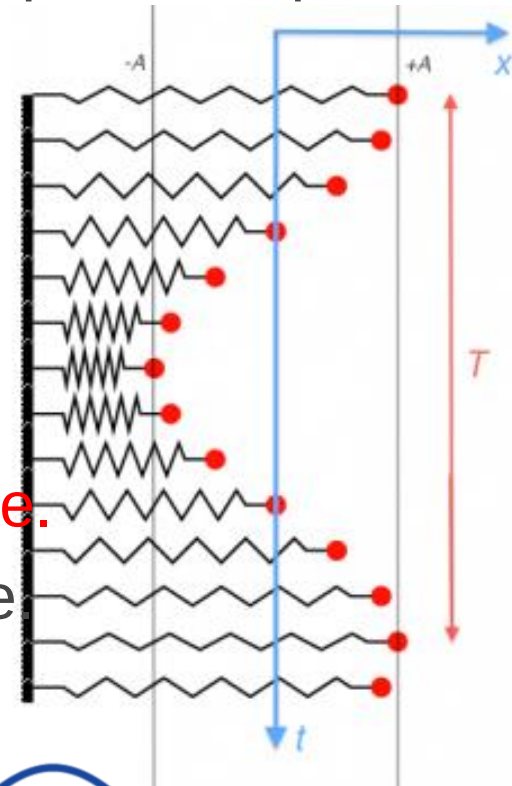


Ans.: e

Problem

Consider the graphical representation of SHM as described mathematically by the equation we have just looked at. When the particle is at point A, what can you say about the particle's position and velocity?

- a) The position and velocity are both positive.
- b) The position and velocity are both negative.
- c) The position is positive and velocity is zero.
- d) The position is negative and velocity is zero.
- e) The position is negative and velocity is positive.
- f) The position is positive and velocity is negative.



Ans.: e

Simple Harmonic Motion (SHM)

$$x(t) = A \cos(\omega t + \phi)$$



- ▶ Motion repeats after one period ($t + T$) – this is when the phase increases by 2π radians.

$$[\omega(t + T) + \phi] - (\omega t + \phi) = 2\pi$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$\omega = 2\pi f$$

Period

Frequency

Simple Harmonic Motion (SHM)

- **Velocity of SHM**

- $v(t) = \frac{dx}{dt}$

- $= \frac{d}{dt} [A \cos(\omega t + \phi)]$

- **Velocity amplitude:**

$$v(t) = -\omega A \sin(\omega t + \phi)$$

- $v_{\max} = \omega A = \sqrt{\frac{k}{m}} A$

- **Acceleration of SHM**

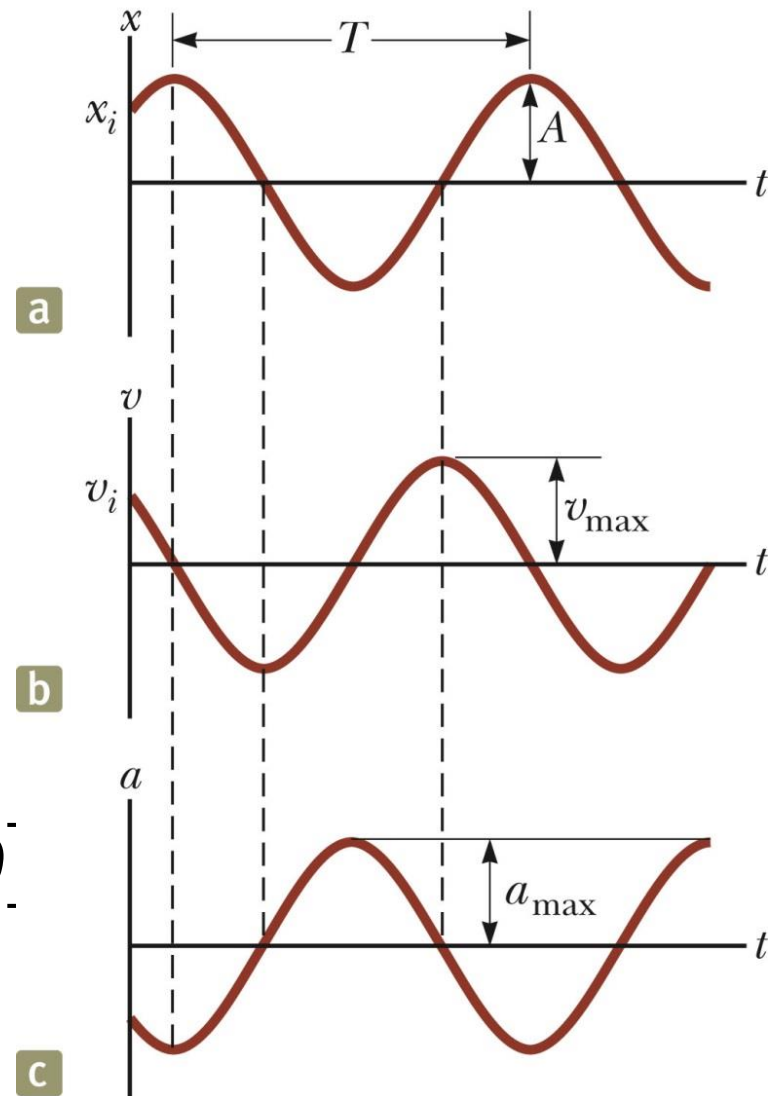
- $a(t) = \frac{dv}{dt} = \frac{d}{dt} [-\omega A \sin(\omega t + \phi)]$

$$a(t) = -\omega^2 A \cos(\omega t + \phi) = -\omega^2 x$$

- **Acceleration amplitude:**

$$a_{\max} = \omega^2 A = \frac{k}{m} A$$

- In SHM $a \propto x$ but opposite in sign, and related by ω^2 .



Example 1

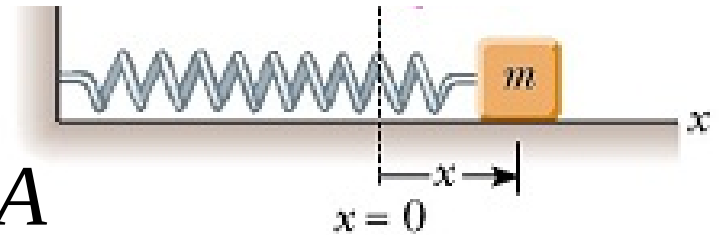
- ▶ A block is set into motion by pulling it from equilibrium by a distance A and releasing it from rest at $t = 0$. **The solutions for the equations for $x(t)$ and $v(t)$ must reflect the initial conditions (BCs).**

$$x(t) = A \cos(\omega t + \phi)$$

$$x(0) = A \cos[\omega(0) + \phi] = A$$

$$v(0) = -\omega A \sin[\omega(0) + \phi] = 0$$

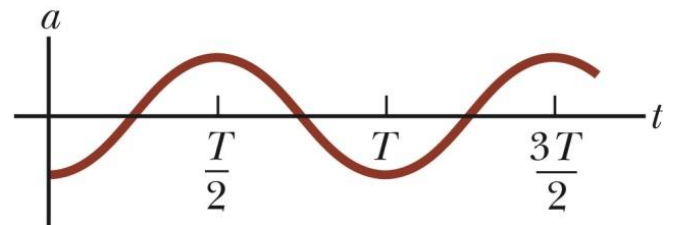
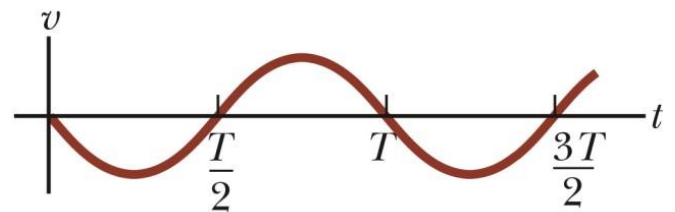
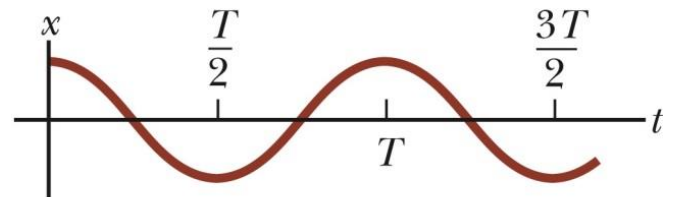
(a)



$$t_i = 0$$

$$x_i = A$$

$$v_i = 0$$



This is satisfied when

$$\cos(\omega(0) + \phi) = 1,$$

therefore $\phi = 0$.

$$\therefore x(t) = A \cos(\omega t)$$

Example 2

Consider the same system as previously, but now it is already oscillating, and we define $t = 0$ as the block passes through the unstretched position ($x = 0$) while moving to the right.

$$x(t) = A \cos(\omega t + \phi)$$

$$x(0) = A \cos[\omega(0) + \phi] = 0$$

$$v(0) = -\omega A \sin[\omega(0) + \phi] = v_i$$

This is satisfied when: $\cos(\phi) = 0$
therefore $\phi = \pm \pi/2$

The second condition is that the velocity and amplitude v_{max} must be positive.

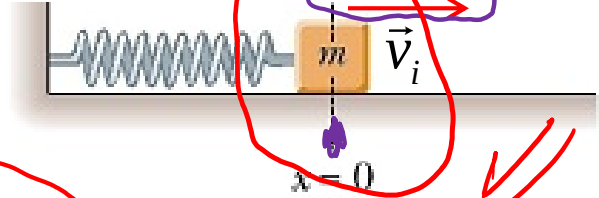
$$v_{max} = v_i = -\omega A \text{ must be } +.$$

$$\text{So, } -\omega A \sin(\phi = -\pi/2) = -\omega A(-1)$$

$$\Rightarrow \phi = -\pi/2 \text{ and } A = v_i / \omega$$

$$\therefore x(t) = \frac{v_i}{\omega} \cos\left(\omega t - \frac{\pi}{2}\right)$$

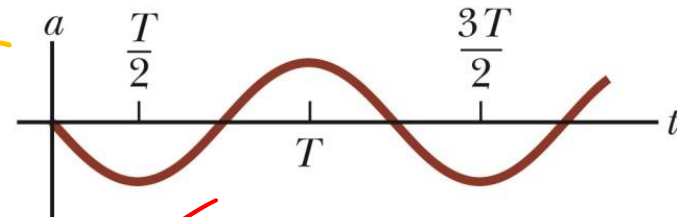
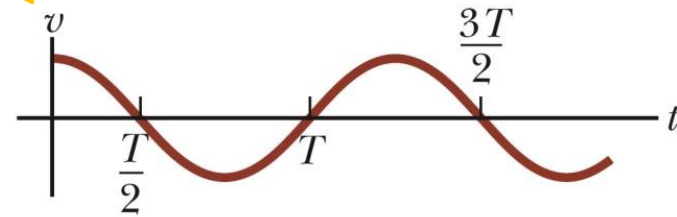
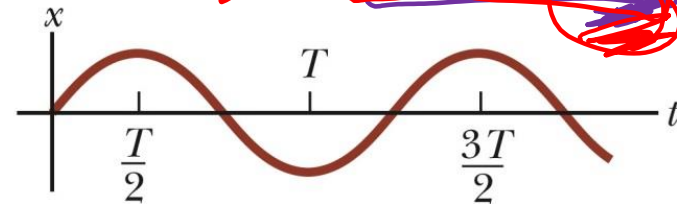
(b)



$$t_i = 0$$

$$x(0) = 0$$

$$v(0) = v_i$$



b

Problem

A cork floating on water undergoes simple harmonic motion. The peak-to-peak distance is 4.0 cm. $T = 1.0$ s. At $t = 0$ the cork is at its minimum height. What are the height and velocity after 10.5 periods?

At $t = 0$

$$x_0 = -2 \text{ cm}$$

$$x_m = 2 \text{ cm}$$

$$\phi = ?$$

$$x(t) = A \cos(\omega t + \phi)$$

$t = 0$

(Evaluate at $t = 0$):

$$\omega = \frac{2\pi}{T}$$

Definition

$$x(t = 0) = -A = A \cos(\omega(0) + \phi)$$

$$\cos(\omega(0) + \phi) = -1$$

$$\phi = \pi$$

Q: When $t = 10.5T = 10.5$ s:

$x = ?$ $v = ?$

$$x = A \cos(\omega t + \phi)$$

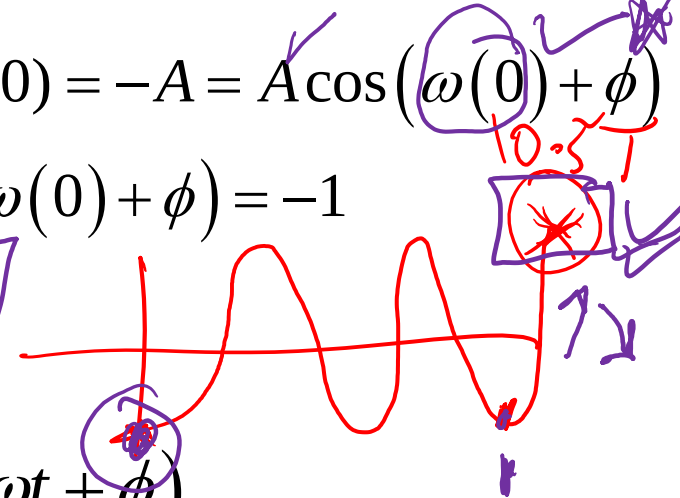
$$x = 2 \cos\left(\frac{2\pi}{T} 10.5T + \pi\right)$$

$$x = 2 \text{ cm}$$

$$v = -\omega A \sin(\omega t + \phi)$$

$$v = -\left(\frac{2\pi}{T}\right) 2 \sin\left(\frac{2\pi}{T} 10.5T + \pi\right)$$

$$v = 0 \text{ cm/s}$$



Problem

$$x(t) = A \cos(\omega t + \phi)$$

A body oscillates with simple harmonic motion according to the equation $x = (6.0 \text{ m}) \cos[(3\pi \text{ rad/s})t + \pi/3 \text{ rad}]$

At $t = 2.0 \text{ s}$, what are the (a) displacement, (b) velocity, (c) acceleration, and (d) the phase of motion? Also, what are the (e) frequency and (f) the period of the motion?

$$(a) \quad x(2) = (6) \cos[(3\pi \times 2) + \pi/3] = 3 \text{ m}$$

$$(d) \quad \omega t + \phi = (3\pi)(2) + \pi/3$$

$$(b) \quad v(2) = -(3\pi)(6) \sin[(3\pi)(2) + \pi/3] = -49 \text{ m/s}$$

$$= 19.9 \text{ rad}$$

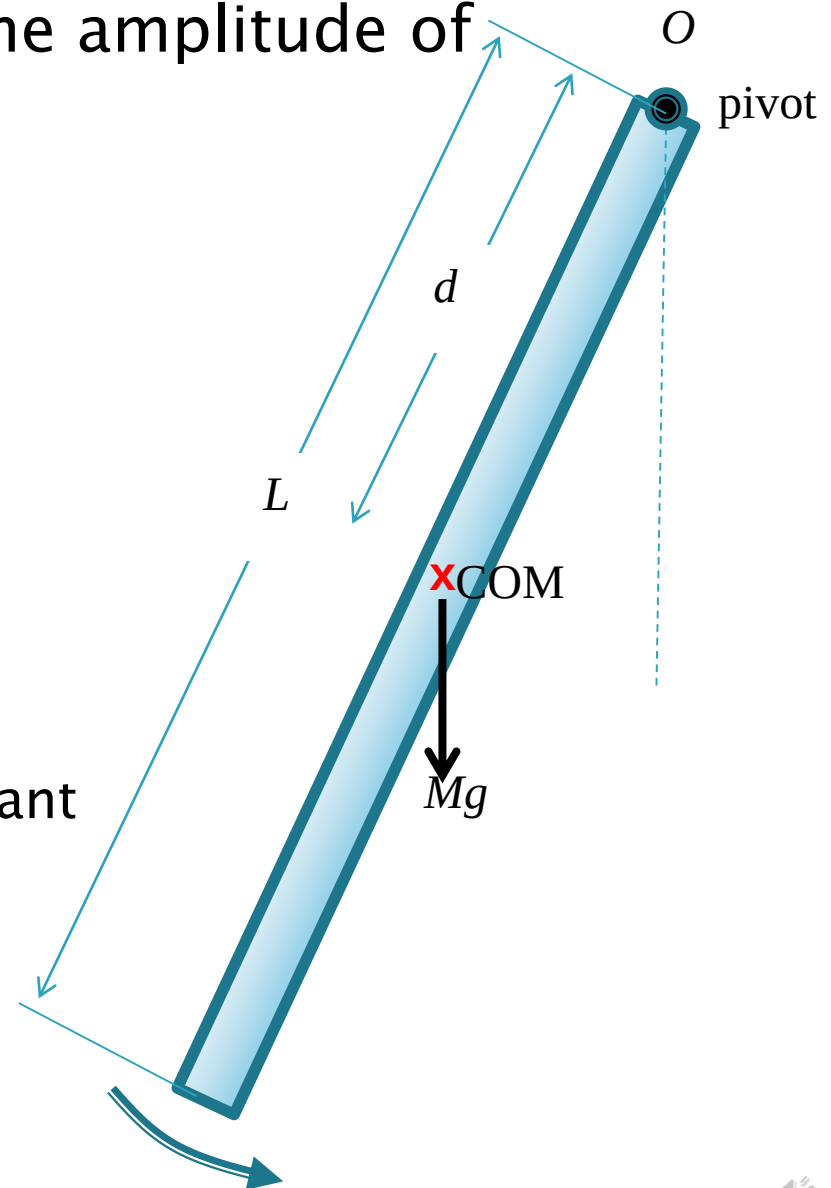
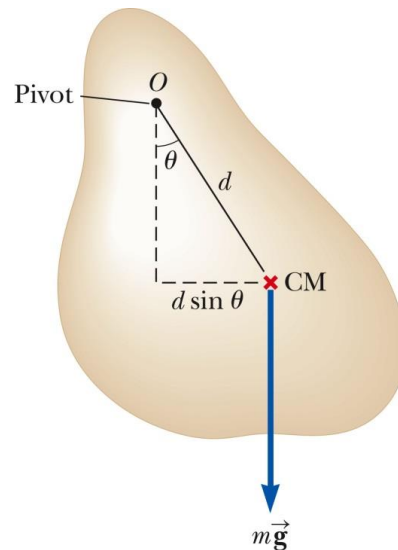
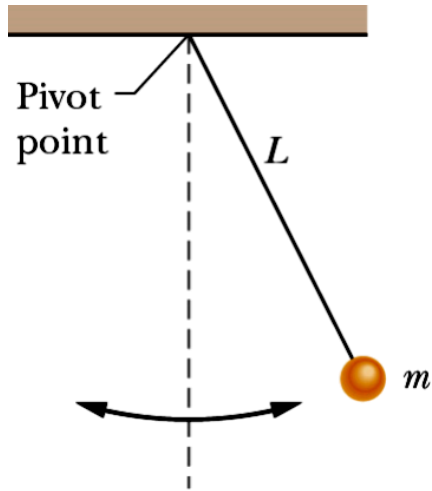
$$(e) \quad f = \frac{\omega}{2\pi} = \frac{3\pi}{2\pi} = 1.5 \text{ Hz}$$

$$(c) \quad a(2) = -\omega^2 x(2) = -(3\pi)^2 (3) = -266.5 \text{ m/s}^2$$

$$(f) \quad T = 1/f = 1/1.5 = 0.67 \text{ s}$$

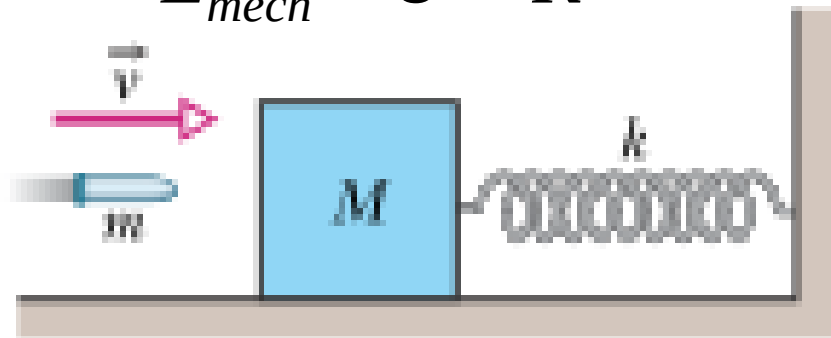
Coming Highlights!

Find the period of oscillation if the amplitude of the motion is small.



Mechanical energy of oscillator is constant

$$E_{mech} = U + K$$





FSK116 – Announcement

Energy in SHM


$$U_s = \frac{1}{2} k x^2 \quad x(t)$$

- ▶ Potential energy of a linear oscillator (spring) depends on how far the spring is stretched out. i.e. Depends on $x(t)$.

$$U(t) = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \varphi)$$

- ▶ Kinetic energy of the system is associated entirely with the mass and depends on its speed $v(t)$. $v(t) = -\omega A \sin(\omega t + \varphi)$

$$K(t) = \frac{1}{2} m v^2 = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \varphi)$$

$$K(t) = \frac{1}{2} k A^2 \sin^2(\omega t + \varphi)$$

Since

$$\omega^2 = \frac{k}{m}$$

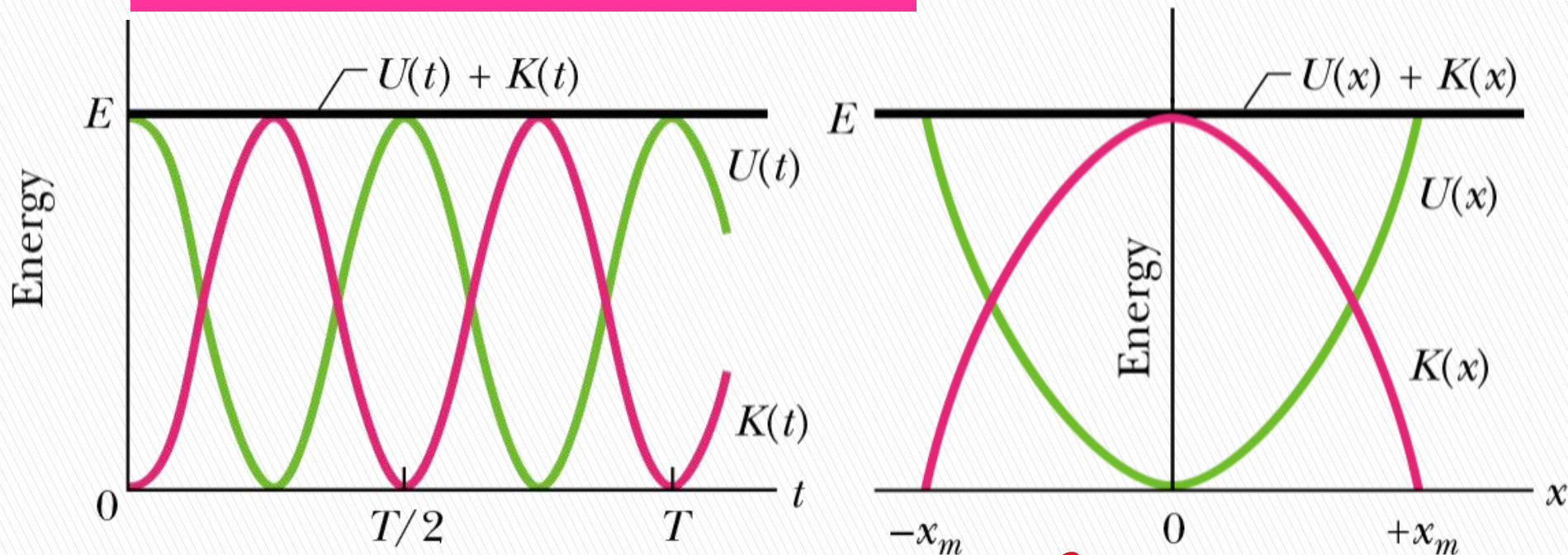
- ▶ Mechanical energy of oscillator. E constant.

$$E = U + K = \frac{1}{2} k A^2$$

Energy in SHM

$$U(t) = \frac{1}{2} kx^2 = \frac{1}{2} kA^2 \cos^2(\omega t + \varphi)$$

$$K(t) = \frac{1}{2} kA^2 \sin^2(\omega t + \varphi)$$



$$E = U + K = \frac{1}{2}kA^2$$

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$$
$$\Rightarrow v^2 = \frac{2}{m} \left(\frac{1}{2}kA^2 - \frac{1}{2}kx^2 \right)$$

Energy in Simple Harmonic Motion

Mechanical energy of the oscillator is constant

$$E_{\text{mech}} = U + K$$

$$K = \frac{1}{2}mv^2 =$$

$$E = \frac{1}{2}kA^2 = U + K$$

$$= \frac{1}{2}kx^2 + \frac{1}{2}mv^2$$

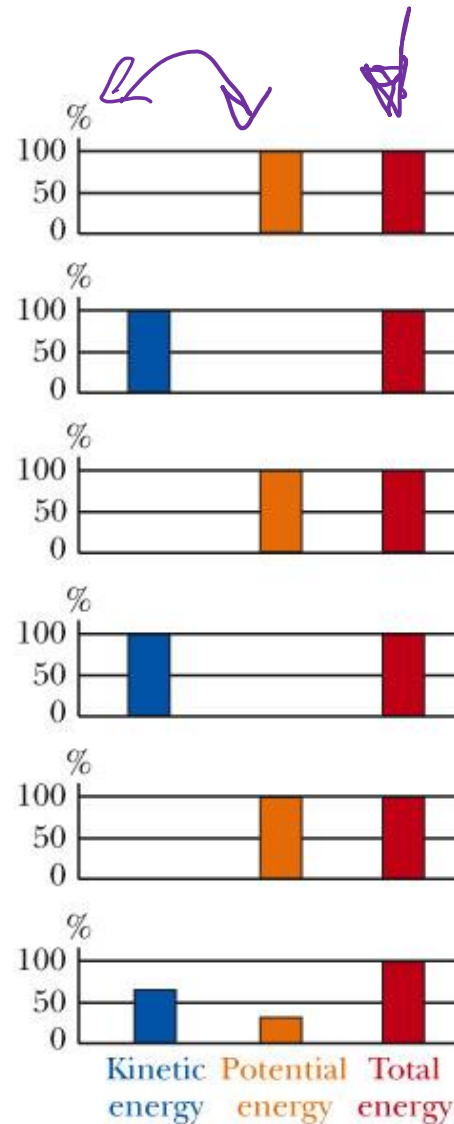
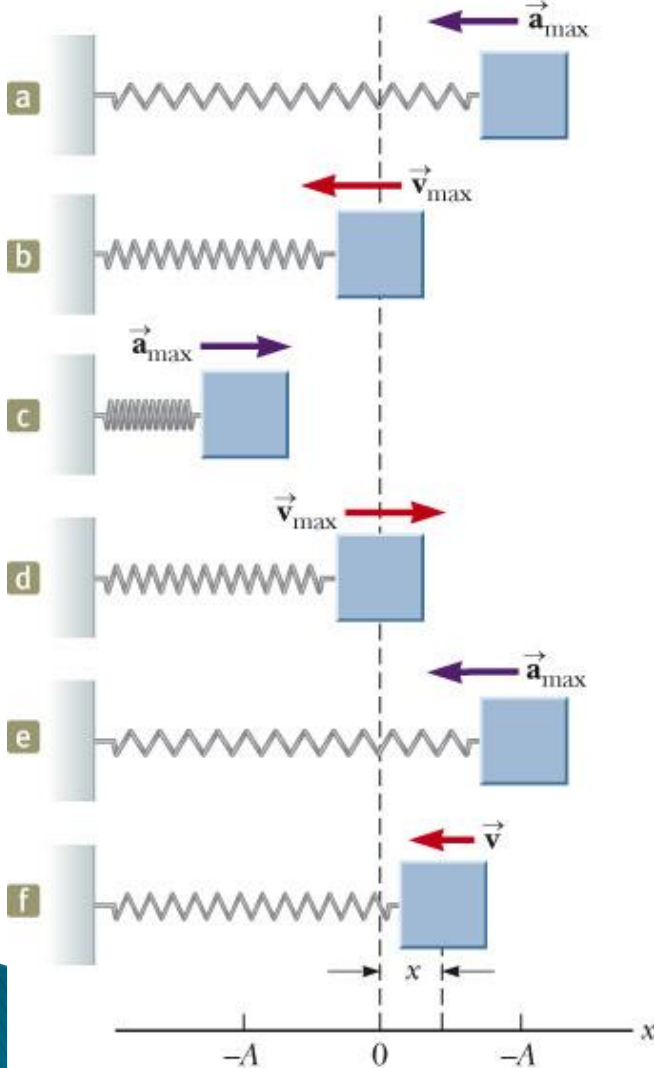
$$E = \frac{1}{2}kA^2$$

$$\Rightarrow v = \pm \sqrt{\frac{k}{m}(A^2 - x^2)} = \pm \omega \sqrt{A^2 - x^2}$$

Speed at any position x

Energy in simple harmonic motion

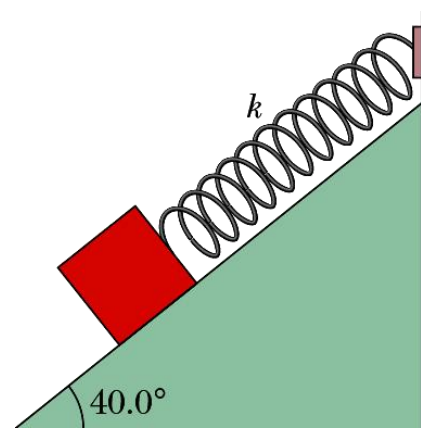
SS



t	x	v	a	K	U
0	A	0	$-\omega^2 A$	0	$\frac{1}{2}kA^2$
$\frac{T}{4}$	0	$-\omega A$	0	$\frac{1}{2}kA^2$	0
$\frac{T}{2}$	-A	0	$\omega^2 A$	0	$\frac{1}{2}kA^2$
$\frac{3T}{4}$	0	ωA	0	$\frac{1}{2}kA^2$	0
T	A	0	$-\omega^2 A$	0	$\frac{1}{2}kA^2$
t	x	v	$-\omega^2 x$	$\frac{1}{2}mv^2$	$\frac{1}{2}kx^2$

Problem

A block weighing 14.0 N, which can slide without friction on an incline at an angle $\theta = 40.0^\circ$, is connected to the top of the incline by a massless spring of un-stretched length 0.450 m and spring constant 120 N/m.



(a) How far from the top of the incline is the block's equilibrium position?

(b) If the block is pulled slightly down the incline and released, what is the period of the resulting oscillations?

(a) $\sum F_x = ma_x \Rightarrow k|x| - mg \sin \theta = 0$

$$mg \sin \theta = k|x_0|$$

$$\therefore |x_0| = \frac{mg \sin \theta}{k}$$

$$= \frac{(14) \sin 40^\circ}{120} = 7.5 \times 10^{-2} \text{ m}$$

$$\text{Equilibrium position} = 0.450 + 0.075 = 0.525 \text{ m}$$

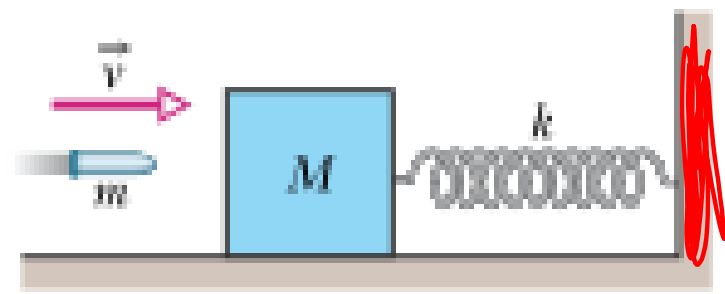
W = mg

(b) $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{W/g}{k}}$

$$= 2\pi \sqrt{\frac{14}{(9.8)(120)}}$$
$$= 0.686 \text{ s}$$



Problem

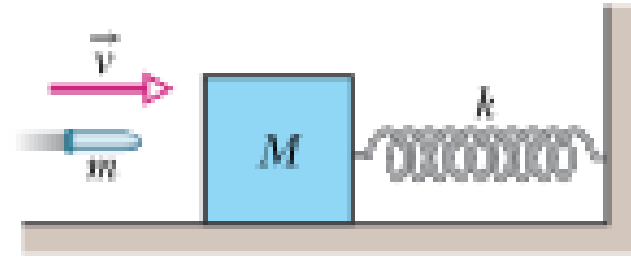


A block of mass $M = 5.4 \text{ kg}$, at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant $k = 6000 \text{ N/m}$. A bullet of mass $m = 9.5 \text{ g}$ and velocity of *magnitude 630 m/s strikes and is embedded in the block* – see Figure. Assuming the compression of the spring is negligible until the bullet is embedded, determine:

- (a) the speed of the block *immediately* after the collision and
- (b) the amplitude of the resulting simple harmonic motion.
- (c) What fraction of the original kinetic energy of the bullet appears as mechanical energy in the harmonic oscillator?

Problem

A block of mass $M = 5.4 \text{ kg}$, at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant $k = 6000 \text{ N/m}$. A bullet of mass $m = 9.5 \text{ g}$ and velocity of magnitude 630 m/s strikes and is embedded in the block – see Figure. Assuming the compression of the spring is negligible until the bullet is embedded, determine:



- (a) the speed of the block immediately after the collision and
- (b) the amplitude of the resulting simple harmonic motion.
- (c) What fraction of the original kinetic energy of the bullet appears as mechanical energy in the harmonic oscillator?

Conservation of momentum for collision and conservation of energy for the oscillator.

$$\begin{aligned} M &= 5.4 \text{ kg} \\ m &= 9.5 \text{ g} = 0.0095 \text{ kg} \\ v_m &= 630 \text{ m/s} \\ k &= 6000 \text{ N/m} \end{aligned}$$

System: Bullet + Block
Isolated

From momentum conservation

$$(a) \quad \underline{mv_i} = (m + M)v_f$$

$$\underline{v_f} = \frac{mv_i}{(m + M)} = \frac{(0.0095)(630)}{(0.0095 + 5.4)} = 1.11 \text{ m/s}$$

Problem cont.

$$E = \cancel{K} + U$$

$$\begin{aligned} M &= 5.4 \text{ kg} \\ m &= 9.5 \text{ g} = 0.0095 \text{ kg} \\ v_m &= 630 \text{ m/s} \\ k &= 6000 \text{ N/m} \end{aligned}$$

(b) **After collision:** $K = \frac{1}{2}(m + M)v_f^2$
At compressed position: $E_{\text{mech}} = U_{s,\text{max}} = \frac{1}{2}k A^2$

$$\Delta E_{\text{mech}} = 0 \Rightarrow (K + U)_i = (K + U)_f$$

$$\frac{1}{2}(m + M)v^2 + 0 = 0 + \frac{1}{2}k A^2$$

$$\begin{aligned} A &= \sqrt{\frac{(m + M)v^2}{k}} = \sqrt{\frac{(5.4095)(1.11)^2}{6000}} \\ &= 0.033 \text{ m} \end{aligned}$$

Original Kinetic Energy of bullet

$$\begin{aligned} \text{(c)} \quad K_{b,i} &= \frac{1}{2}mv_{b,i}^2 = \frac{1}{2}(0.0095)(630)^2 \\ &= 1.89 \times 10^3 \text{ J} \end{aligned}$$

Kinetic Energy of harmonic oscillator

$$\begin{aligned} K &= \frac{1}{2}(m + M)v^2 = \frac{1}{2}(5.4095)(1.11)^2 \\ &= 3.33 \text{ J} \end{aligned}$$

$$\frac{K}{K_{b,i}} = \frac{3.33 \text{ J}}{1.89 \times 10^3} = 0.00176 \approx 0.176\%$$

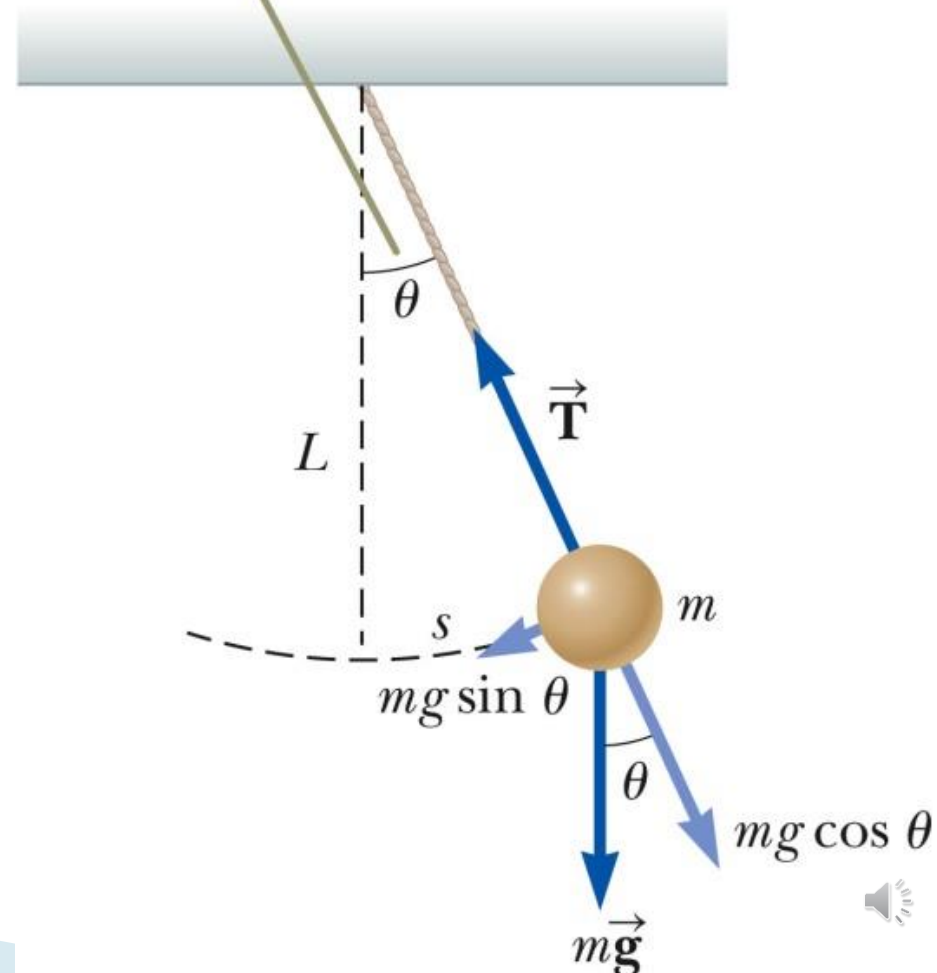
What happened to the rest of the energy?

SHM & circular motion

- ▶ Read Section 15-4

Simple Pendulum

When θ is small, a simple pendulum's motion can be modeled as simple harmonic motion about the equilibrium position $\theta = 0$.



Pendulums – Simple Pendulum

- ▶ **Simple pendulum**: mass m suspended from an unstretchable, (light) massless string of length L .

- ▶ Forces on mass:

- weight mg and tension T .

- ▶ tangential component: $mg \sin \theta$

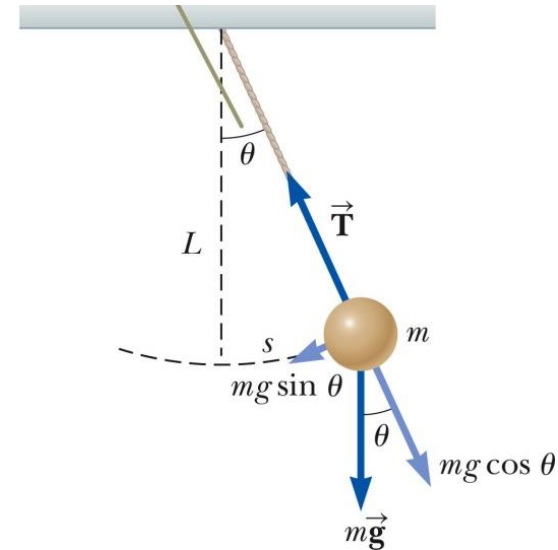
radial component: $mg \cos \theta$

- ▶ **Tangential component produces restoring force.**

ie. brings mass back to equilibrium position $\theta = 0^\circ$.

$$F_t = ma_t \rightarrow -mg \sin \theta = m \frac{d^2 s}{dt^2} \Rightarrow -g \sin \theta = \frac{d^2 s}{dt^2}$$

- ▶ Tangential acceleration i.t.o. position s (sector)



Pendulums – Simple Pendulum

▶ $s = L\theta$ and L is constant for pendulum

▶
$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta$$

▶ Compare to $\frac{d^2x}{dt^2} = -\frac{k}{m}x$, the criteria for SHM.

▶ If we assume small angles: $\theta < 10^\circ$

▶ Then $\sin\theta \approx \theta$ in radians (See Table 15.1)

▶ and
$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

▶ Then $\theta = \theta_{\max} \cos(\omega t + \phi)$ is a solution to the diff. eqn.

$$\omega = \sqrt{\frac{g}{L}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L}{g}}$$

A Required Derivation!

Pendulums – Simple Pendulum

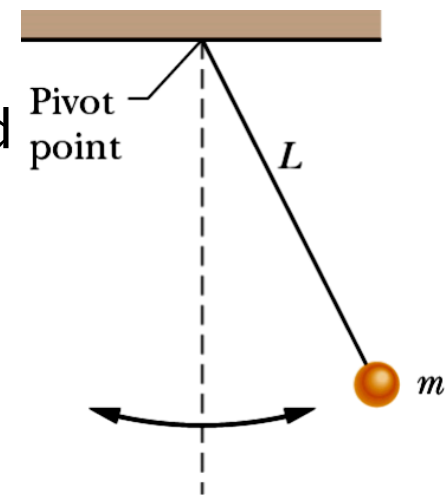
- ▶ $T = 2\pi\sqrt{\frac{L}{g}}$
- ▶ Period is ONLY dependent on the length of the string and g .
- ▶ Simple pendulum used in old grandfather clocks.

Problem

A pendulum has a length of 2 m. The mass is displaced from the vertical position by giving it a small push.

The maximum speed is 0.11 m/s.

What is the maximum horizontal displacement?



$$\theta = \theta_m \cos(\omega t + \delta)$$

$$x = \theta L \quad \text{where } s = x \text{ and are very small}$$

$$x_m = \theta_m L$$

We also know that:

$$\frac{x}{L} = \frac{x_m}{L} \cos(\omega t + \delta)$$

$$x = x_m \cos(\omega t + \delta)$$

$$\omega = \sqrt{\frac{g}{L}} = \sqrt{\frac{9.8 \text{ m/s}^2}{2 \text{ m}}} = 2.21 \text{ rad/s}$$

$$\text{But: } v = \frac{dx}{dt} = -\omega x_m \sin(\omega t + \delta)$$

$$\omega x_m = 0.11 \text{ m/s} \quad \text{DFN of velocity amplitude}$$

$$\therefore x_m = \frac{0.11 \text{ m/s}}{\omega} = 0.05 \text{ m}$$

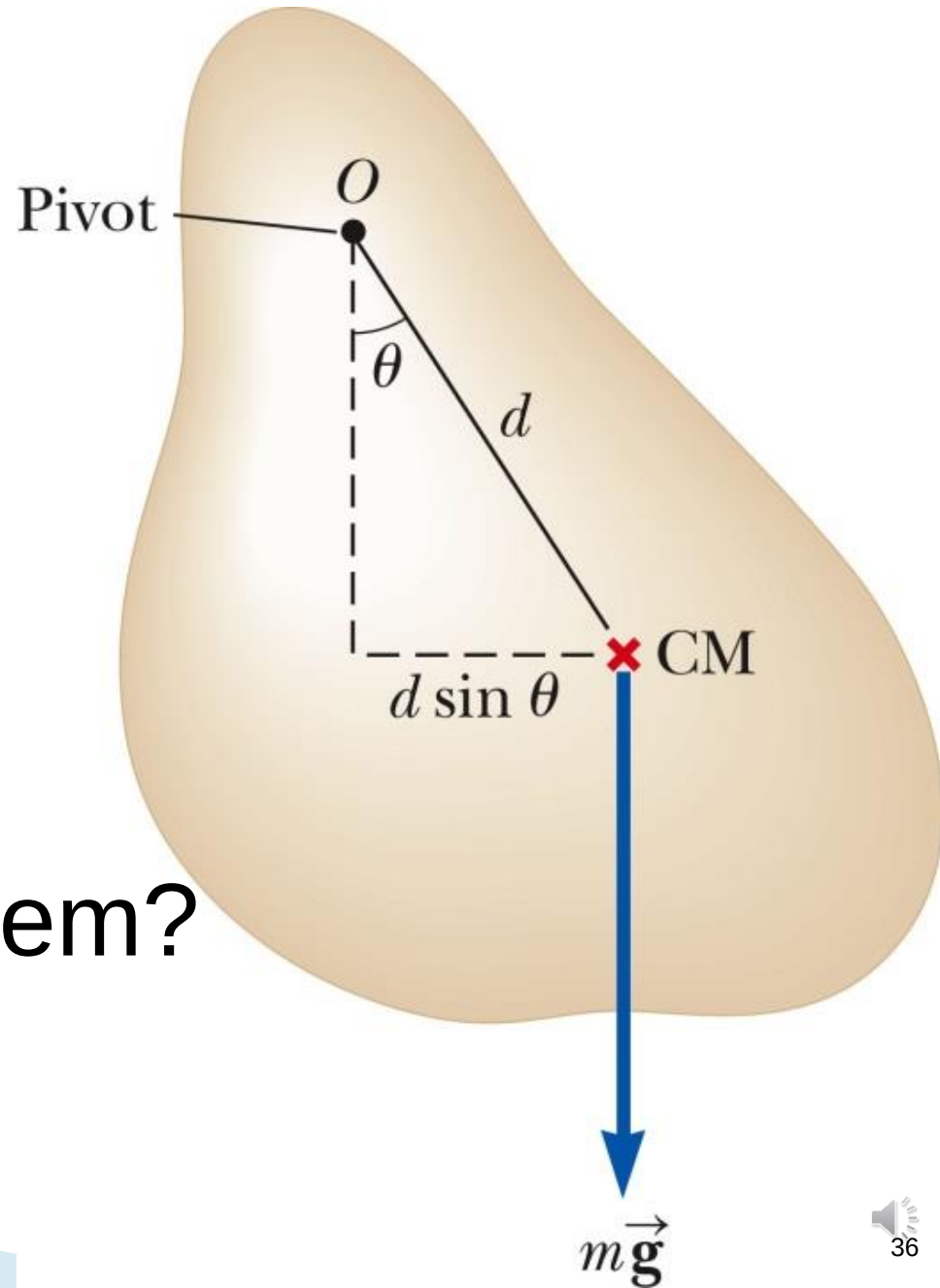
Pendulums – Physical Pendulum

$$\tau = -mgd \sin \theta$$

$$\sum \tau_{net} = I\alpha$$

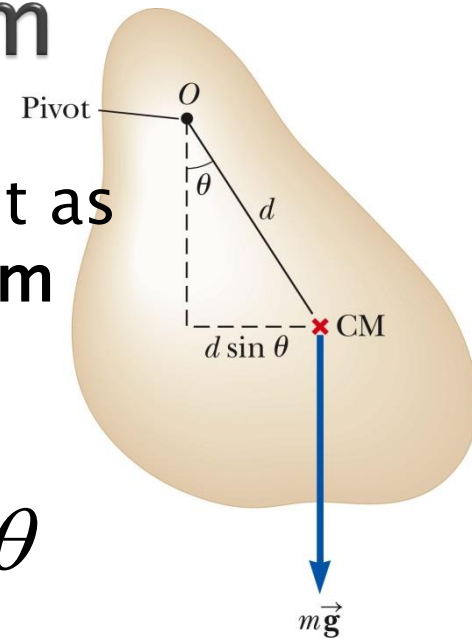
REMEMBER

Parallel Axis Theorem?



Pendulums – Physical Pendulum

- ▶ If an object oscillates about a fixed axis not passing through COM – WE CAN'T treat it as a simple pendulum – it is a **Physical Pendulum**
- ▶ Consider: Rigid object rotating about O .
- ▶ Gravitational force → gives rise to (-) torque



$$\sum \tau_{net} = I\alpha \quad \tau = -mgd \sin \theta$$
$$-mgd \sin \theta = I \frac{d^2 \theta}{dt^2}$$

- ▶ For small angles: $\frac{d^2 \theta}{dt^2} = -\frac{mgd}{I} \theta = -\omega^2 \theta \Rightarrow \omega = \sqrt{\frac{mgd}{I}}$

- ▶ So $\theta = \theta_{max} \cos(\omega t + \phi)$ **Is a solution of the Diff Eqn.**

$$\text{From } \omega = \sqrt{\frac{mgd}{I}} \Rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{mgd}}$$

Example 15.6

A uniform rod of mass M and length L is pivoted about one end and oscillates in a vertical plane. Find the period of oscillation if the amplitude of the motion is small.

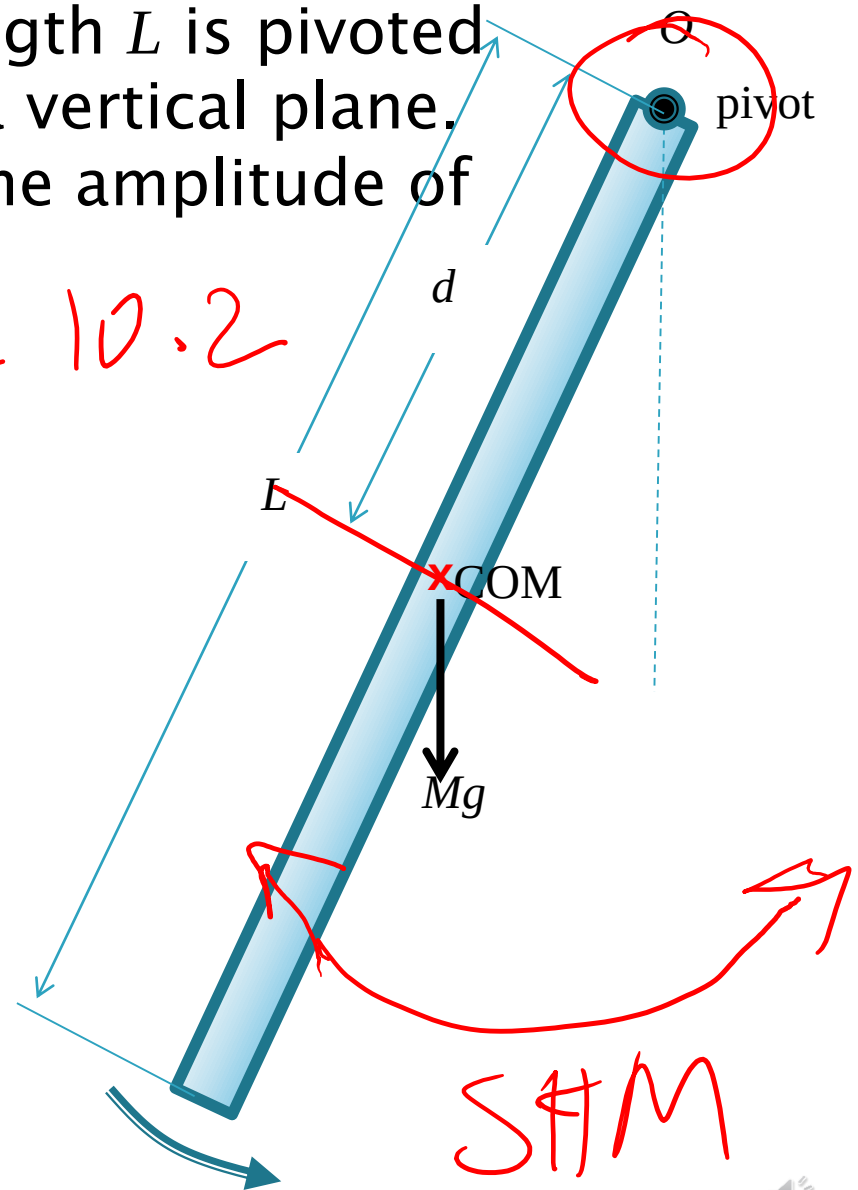
$$I = \frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2 = \frac{1}{3}ML^2$$

Table 10.2

$$(a) \quad T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$T = 2\pi \sqrt{\frac{(1/3)ML^2}{Mg(L/2)}}$$

$$= 2\pi \sqrt{\frac{2L}{3g}}$$



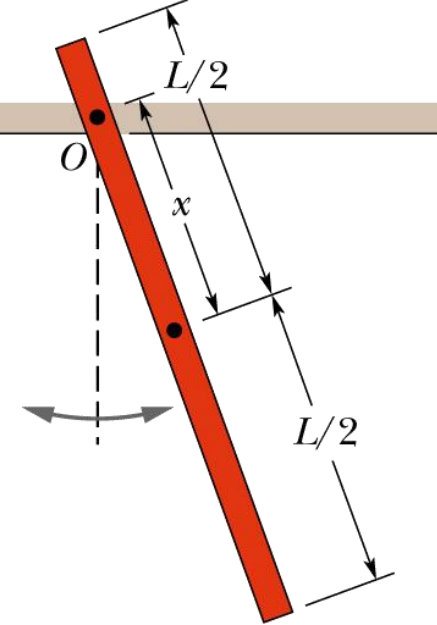
Problem

A stick of length $L = 1.00$ m oscillates as a physical pendulum.

(a) Derive an equation for the period of the pendulum in terms of L and x the distance from the pivot point to the centre of mass.

(b) For what value of x/L is the period a minimum?

(c) What is this minimum period?



$$(a) \quad T = 2\pi \sqrt{\frac{I_0}{mgx}}$$

$$(b) \quad \text{Let } \frac{dT}{dx} = \frac{d}{dx} \left(\frac{L^2}{12gx} + \frac{x}{g} \right) = 0$$

$$\begin{aligned} I_0 &= I_{com} + mh^2 \\ &= \frac{mL^2}{12} + mx^2 \end{aligned}$$

$$\left(\frac{-L^2}{12gx^2} + \frac{1}{g} \right) = 0$$

$$\frac{x^2}{L^2} = \frac{1}{12}$$

$$\frac{x}{L} = \sqrt{\frac{1}{12}} = 0.289$$

$$\begin{aligned} (c) \quad x/L &= 0.289 \\ \Rightarrow x &= 0.289 \times L \\ &= 0.289 \times 1 \text{ m} \\ &= 0.289 \text{ m} \end{aligned}$$

$$\begin{aligned} T &= 2\pi \sqrt{\frac{(mL^2/12) + mx^2}{mgx}} \\ &= 2\pi \sqrt{\frac{L^2 + 12x^2}{12gx}} \end{aligned}$$

$$\begin{aligned} T_{\min} &= 2\pi \sqrt{\frac{(1 \text{ m})^2 + 12(0.289 \text{ m})^2}{12(9.8 \text{ m/s}^2)(0.289 \text{ m})}} \\ &= 1.53 \text{ s} \end{aligned}$$

Problems

- ▶ Make sure you do all the Examples and Problems from the sections covered in this chapter ie. in these notes.

▶ **THE END!**